Online Book Store Sales Analysis

```
CREATE TABLE Books (
  Book_ID SERIAL PRIMARY KEY,
  Title VARCHAR(100),
  Author VARCHAR(100),
  Genre VARCHAR(50),
  Published_Year INT,
  Price NUMERIC(10, 2),
  Stock INT
);
```

```
CREATE TABLE Customers (
  Customer_ID SERIAL PRIMARY KEY,
  Name VARCHAR(100),
  Email VARCHAR(100),
  Phone VARCHAR(15),
  City VARCHAR(50),
  Country VARCHAR(150)
);
```

```
CREATE TABLE Orders (
  Order_ID SERIAL PRIMARY KEY,
  Customer_ID INT REFERENCES Customers(Customer_ID),
  Book_ID INT REFERENCES Books(Book_ID),
  Order_Date DATE,
  Quantity INT,
  Total_Amount NUMERIC(10, 2)
);
```

```sql
SELECT * FROM Books;

SELECT * FROM Customers;

SELECT * FROM Orders;
```

```sql
-- 1) Retrieve all books in the "Fiction" genre:

Select title,genre from Books
where genre='Fiction';
```

```sql
-- 2) Find books published after the year 1950:
select title,published_year from Books
where published_year >1950;
```

```sql
-- 3) List all customers from the Canada:
Select name,country from Customers
where country='Canada';
```

```sql
-- 4) Show orders placed in November 2023:
Select * from Orders
where Order_date between '2023-11-01' and '2023-11-30';
```

```sql
-- 5) Retrieve the total stock of books available:

Select Sum(stock) as Total_Stock from Books;
```

```sql
-- 6) Find the details of the most expensive book:
select * from Books
```

```sql
order by price desc limit 1;
```

```sql
-- 7) Show all customers who ordered more than 1 quantity of a book:
Select * from Orders
where quantity >1 order by quantity asc;
```

```sql
-- 8) Retrieve all orders where the total amount exceeds $20:
Select * from Orders
where total_amount >20 order by total_amount asc;
```

```sql
-- 9) List all genres available in the Books table:
Select Distinct(genre) from Books;
```

```sql
-- 10) Find the book with the lowest stock:
Select * from Books
order by stock asc limit 1;
```

```sql
-- 11) Calculate the total revenue generated from all orders:
Select sum(total_amount) as Total_Revenue from Orders;
```

```sql
-- Advance Questions :

-- 1) Retrieve the total number of books sold for each genre:
Select b.genre, sum(o.quantity) as Total_book_Sold
from Orders o
join
Books b
on b.book_id=o.book_id
```

```sql
group by b.genre;
```

```sql
-- 2) Find the average price of books in the "Fantasy" genre:
Select round(Avg(price),2) as Avg_Price from Books
where genre='Fantasy';
```

```sql
-- 3) List customers who have placed at least 2 orders:

select o.customer_id,c.name, count(o.order_id) Total_order_per_customer
from Orders o
join
Customers c
on c.customer_id=o.customer_id
group by o.customer_id,c.name
having count(order_id) >=2 order by Total_order_per_customer asc;
```

```sql
-- 4) Find the most frequently ordered book:
Select o.book_id,b.title, count(o.order_id) as Most_frequent_order_book
from Orders o
join
Books b
on
b.book_id=o.book_id
group by o.book_id,b.title
order by Most_frequent_order_book desc limit 1;
```

```sql
-- 5) Show the top 3 most expensive books of 'Fantasy' Genre :
```

select title,genre,price from Books

where genre='Fantasy'

order by price desc limit 3;

-- 6) Retrieve the total quantity of books sold by each author:

select b.author,sum(o.quantity) as Total_quantity

from Books b

join Orders o

on b.book_id=o.book_id

group by b.author;

-- 7) List the cities where customers who spent over $30 are located:

select c.customer_id,c.name,c.city,c.country,sum(o.total_amount) as Total_spent

from Customers c

join

Orders o

on

c.customer_id=o.customer_id

where total_amount >30

group by

c.customer_id,c.name,c.city,c.country

order by Total_spent asc;

-- 8) Find the customer who spent the most on orders:

select c.customer_id,c.name,sum(o.total_amount) as Total_spent

from Customers c

```
join
Orders o
on
c.customer_id=o.customer_id
where total_amount >30
group by
c.customer_id,c.name
order by Total_spent desc limit 1;
```

```
--9) Calculate the stock remaining after fulfilling all orders:
SELECT b.book_id, b.title, b.stock, COALESCE(SUM(o.quantity),0) AS Order_quantity,
        b.stock- COALESCE(SUM(o.quantity),0) AS Remaining_Quantity
FROM books b
LEFT JOIN orders o ON b.book_id=o.book_id
GROUP BY b.book_id ORDER BY b.book_id;
```
